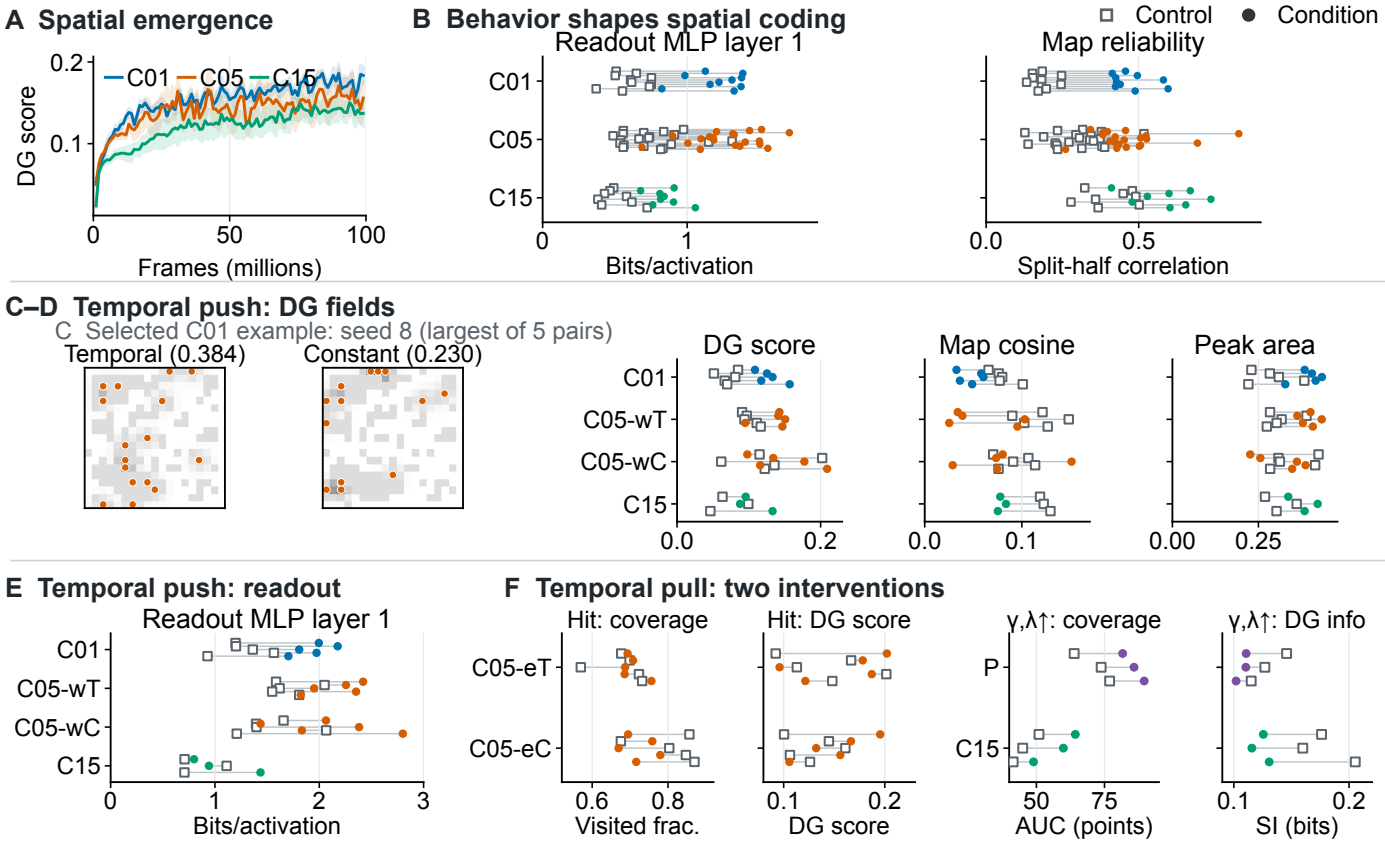

A Spatial emergence
C01
C05
C15
DG score
0.2
0.1
0
50
100
Frames (millions)
B Behavior shapes spatial coding
Control
Condition
Readout MLP layer 1
C01
C05
C15
0
1
Bits/activation
Map reliability
0.0
0.5
Split-half correlation
C–D Temporal push: DG fields
C Selected C01 example: seed 8 (largest of 5 pairs)
Temporal (0.384)
Constant (0.230)
DG score
C01
C05-wT
C05-wC
C15
0.0
0.2
Map cosine
0.0
0.1
Peak area
0.00
0.25
E Temporal push: readout
Readout MLP layer 1
C01
C05-wT
C05-wC
C15
0
1
2
3
Bits/activation
F Temporal pull: two interventions
Hit: coverage
C05-eT
C05-eC
0.6
0.8
Visited frac.
Hit: DG score
0.1
0.2
DG score
γ,λ↑: coverage
P
C15
50
75
AUC (points)
γ,λ↑: DG info
0.1
0.2
SI (bits)